liquid-cooled battery packs and are testing the same for all upcoming electric two-wheelers, including the mass segment electric motorcycle," he says.

For Hop, the slow-speed electric two-wheeler market is also something that will remain consistent in India for years to come. As a matter of fact, out of the ₹14 billion revenue it posted last year, the majority came from slow-speed scooters.

"We believe that slow-speed electric scooters are what help a lot of Indians experience electric mobility for the very first time. These customers will also be our brand ambassadors in the high-speed electric two-wheeler market," Gupta explains.

As per Gupta, senior citizens, women, and teenagers will continue to invest in slow-speed electric scooters. He added that slow-speed models still dominate China's electric two-wheeler market.

Having raised $3 million from external investors, Hop aims to close the current fiscal at revenue north of ₹22 billion. Gupta highlighted that while slow-speed retail revenue will still dominate their overall revenue figures, the share contributed by high-speed motorcycles will increase 'substantially.'

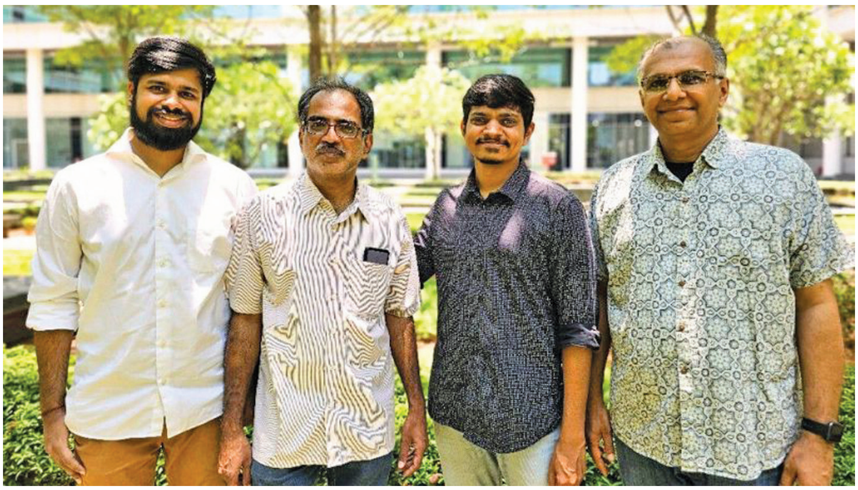
—*Mukul Yudhveer Singh*

2 "WE ASPIRE TO BE THE FIRST CHOICE FOR COMPANIES SEEKING TO TRANSITION TO RISC-V"

Chennai-based startup InCore Semiconductor aims to lead the transition to indigenously designed chips through RISC-V architecture

Identifying itself as a spec-to-silicon company, the fabless startup InCore Semiconductors was founded by GS Madhusudan (CEO), Neel Gala (CTO), Gautam Doshi (Chief Architect), and Arjun Menon (Chief Engineer) in 2018. Specialising in processor IP design in the RISC-V architecture, InCore originated from the Shakti project at IIT Madras.

The startup offers customisable processor IP core solutions supporting various technology nodes with tailored solutions for segments such as low-end and high-end IoT, security, automotive, and storage applications. "We partner with companies to provide comprehensive chip design services. We contribute our IP and high-level architecture while our partners handle the rest of the process," says Madhu.



InCore Co-founders (L to R): Arjun Menon (Chief Engineer), G.S. Madhusudan (CEO), Neel Gala (CTO), and Gautam Doshi (Chief Architect)

InCore offers two RISC-V processor IP cores—a two-stage embedded core Azurite for low-power applications and Calcite for higher performance. "Azurite is designed to supplant ARM processors like M0 to M4 in specific ultra-low power consumption applications, including industrial controllers, smart cards, energy meters, and select automotive systems (excluding ADAS). Calcite boasts greater Linux capability and is well-suited for applications demanding higher performance, such as handheld devices, factory automation, networking on-chip, and storage controllers, achieving speeds of up to 800 megahertz," explains CTO Neel Gala. The cores are being deployed on customer silicon starting early 2024.

The startup uses Bluespec SystemVerilog for hardware development to facilitate parameterisation and scalability in design code. Gala